

PA3 – Tabular Q-Learning (Nim)

Due: May 28, 2026 at 11:59 PM

Overview

In this assignment you will implement tabular Q-learning for Nim, a two-player turn-based strategy game. Player 0 learns from simulated games against simple opponents, then evaluates the learned greedy policy. All graded code tasks are marked with TODO comments in `PA3.ipynb`; Question 4 is a short manual response using the reporting cell near the end of the notebook.

Rules and Submission

- Submit `PA3.ipynb` to Gradescope as an `.ipynb` file, not a PDF.
- Complete all TODOs in Questions 0–3 and fill in the Question 4 responses.
- Do not rename required functions or change function signatures.
- Use only the packages already imported in the notebook. Do not use external RL libraries such as Gymnasium or Stable-Baselines3.
- Optional Question 5, the SARSA extension, may remain unimplemented.
- GenAI tools may be used on assignments, but not tests or exams, if their use is documented and acknowledged honestly.

Point Distribution

Section	Points
Question 0: Nim setup and utilities	15
Question 1: Core Q-learning functions	40
Question 2: Training loop	25
Question 3: Policy and evaluation	15
Question 4: Short written responses	5

Important Notes

- This assignment uses a different Nim variant than the one discussed in class: here the game starts from multiple piles, (1, 3, 5), and the player who takes the last object wins. The class example used one pile starting with 6 objects and the misere rule, where taking the last object loses.

- Reward is always from player 0's perspective: +1 win, -1 loss, 0 otherwise. Player 0 always moves first.
- In Question 2, a nonterminal Q-learning transition includes the agent move and the opponent response. Update the agent's (`state`, `action`) using the state after the opponent moves.
- The Question 2 checkpoint trains `Q_small` with `episodes=200`, `alpha=0.5`, `gamma=0.9`, and `epsilon=0.2`. Use that checkpoint output for Question 4.
- Run the nearby checkpoint cells as you finish each question. All graded cells through Question 4 should run without errors before you submit.
- Gradescope shows some checks immediately; additional hidden checks run during grading.